Holistic Accuracy and the Lockean Thesis

Abstract. Dorst argues that the Lockean thesis follows from the core of accuracy-first epistemology even when the accuracy of a belief state is not the sum of the accuracies of its members. I show that this separability-free derivation is invalid. Its steps hold exactly when accuracy is separable, so it does not reach the holist it addresses. A natural holistic measure, which adds a premium for a belief state with no false member, satisfies all of Dorst's stated premises yet violates the Lockean thesis for every positive premium: rational belief in a proposition can be lost as credence in it rises. Under this measure one believes the sufficiently probable consequences of a working hypothesis chosen for its probability and informativeness. Expected accuracy is therefore neutral between Lockean and guess-like belief. It has a limit, though: no truth-directed value for answers yields the split-invariant belief of normality theories.

1. Introduction

Accuracy-first epistemology holds that rational doxastic states are those that best promote accuracy by the agent's own lights. Applied to outright belief, with true beliefs scored positively and false ones negatively, it yields a Lockean thesis: believe a proposition just in case your credence in it exceeds a threshold. The separable version of this result is due to Easwaran (2016; see Dorst 2019, 7). Dorst (2019) extended it in two directions. Letting the values of true and false belief vary with proposition and context yields variable thresholds (his Corollary 2). And, he argued, dropping Separability, the assumption that a belief state's accuracy is the sum of the accuracies of its members, changes nothing: given a further premise, Sequence-Irrelevance, Lockeanism follows anyway (his Proposition 5). Hence "The connection between epistemic utility theory and the Lockean Thesis is robust" (16), and "Lockeanism looked inevitable" (17).[1]

I argue that it is not robust. My thesis is that whether maximizing expected accuracy yields Lockean belief turns on whether accuracy is separable: Dorst's separability-free derivation is valid only for separable measures, and a natural holistic measure that satisfies all its stated premises yields non-Lockean belief instead. Three results support this. Theorem 1 locates two unstated premises in the derivation and shows that together they amount to Separability. Theorem 2 gives the counterexample: a measure that adds a premium for an error-free belief state meets every premise Dorst states and violates Proposition 5 for every positive premium, because it makes belief non-monotone in credence. Theorem 3 says what that measure delivers instead. Finally, Theorem 4 marks a limit: whatever holism buys, it does not buy the split-invariant belief of knowledge-aimed theories.

The thesis does not say that Lockeanism or Separability is false, that holistic accuracy yields deductively closed belief in general (it need not), or that Dorst's reduction of belief to credence fails (§5).

[1] Page references to Dorst are to the accepted manuscript deposited in the Oxford University Research Archive, the version I was able to read; I could not consult the published pagination, and the published text may differ. Quotations from Rothschild are from the author's preprint dated 1 June 2021.

The closest prior work is Dorst's. Rothschild, after characterizing Lockean belief sets by additive scoring, writes that "Most interesting, I think, would be to see what can be found when the additivity assumption is weakened" (2021 preprint, 15). Wang (2025) obtains non-Lockean belief from expected-utility maximization. According to its abstract, which is all I have been able to read, it concludes that "deductively cogent expected accuracy maximizers are neither Lockean nor Humean." Those maximizers choose among deductively cogent belief states. Dorst's Accuracy First forbids exactly this restriction. He says that deductive closure, imposed from outside, is "the reason—in fact, the only reason—why" Hempel, Levi and Maher "do not end up with Lockeanism" (5). My counterexample needs no such restriction: it keeps the unrestricted option space and every premise Dorst states, and it changes only the value of combinations of beliefs. Holguín (2022, §5) argues that rational thinking is non-monotonic in credence and that sophisticated Lockeanism therefore fails. Theorem 2 shows that accuracy-first epistemology, once it drops Separability, can be non-monotonic too.

The stakes extend beyond formal epistemology. If accuracy alone settled the bridge between credence and belief, the Lockean would win that debate by default. If, as I argue, the shape of the bridge depends on how we value combinations of true beliefs, then the debate over Lockean, closed and guess-like belief becomes a debate about epistemic value, and accuracy-first epistemology is a framework for conducting it rather than a verdict.

2. Dorst's derivation

Let W be a finite set of worlds, let propositions be subsets of W, and let a belief state B be any set of propositions. Accuracy First requires that the options be unrestricted: "there are no external requirements on B—it need not be consistent, or deductively closed, or nonempty" (8). An accuracy measure assigns a real number $A(B, w)$ to each state at each world. Given a probability function C on W, the expected accuracy is $EA_C(B) = \sum_w C(w)A(B, w)$, and Maximize Expected Accuracy says that B is rational only if no state has higher expected accuracy.

Under Separability with proposition-dependent values, $A(B, w) = \sum_{p \in B} v_p(w)$, where $v_p(w) = T_p > 0$ if $w \in p$ and $v_p(w) = F_p < 0$ otherwise. Then $EA_C(B) = \sum_{p \in B} e_p$, where

$$e_p = C(p)T_p + (1 - C(p))F_p.$$

So p belongs to an optimal state iff $e_p \geq 0$, that is, iff $C(p) \geq \tau_p = -F_p/(T_p - F_p)$ (Corollary 2). Dorst also imposes Conservativeness: in the uniform case $\mid T \mid < \mid F \mid$, which makes $\tau > 1/2$.

Section 4.2 grants the holist that A may depend on the current state B_0 "and the value (accuracy) of any given belief-state B cannot be broken down into the value of the various propositions it contains" (15). Dorst assumes that adding a true belief strictly raises accuracy and adding a false one strictly lowers it; call this *truth-directedness*. He then reasons:

> Since A_{B_0} is a real-valued function, there must be some $t > 0$ such that $A_{B_0}(B_1, \mathcal{P}, w) = A_{B_0}(B_0, \mathcal{P}, w) + t$. Write t as $T_p^{B_0}$ (15)

Here $B_1 = B_0 \cup \{p\}$, and a parallel step introduces $F_p^{B_0}$ for false additions. "It's crucial to emphasize that this assumes only that A_{B_0} is well-defined" (15). With these constants, Proposition 4 follows: if beliefs are formed in a sequence ending with p, the final step maximizes expected accuracy only if $C(p)$ lies on the right side of $\tau_p^{B_0} = -F_p^{B_0}/(T_p^{B_0} - F_p^{B_0})$. Then:

> Sequence-Irrelevance: Holding fixed all other factors, rational constraints on beliefs are not affected by the sequence (if any) in which they are formed. (16)

Proposition 5 states that, given Sequence-Irrelevance, "there are belief-independent values T_p and F_p for each proposition such that B is rational only if" $p \in B$ implies $C(p) \geq \tau_p$ and $p \notin B$ implies $C(p) \leq \tau_p$ (16). The proof fixes a sequence that puts p last, takes "any $(n-1)$-stage belief-state B^-", and uses Sequence-Irrelevance to conclude that the threshold relative to B^- constrains "all rational belief-states" (26).

3. The counterexample

Define the *record measure* with premium $\beta > 0$:

$$A_\beta(B, w) = \sum_{p \in B} v_p(w) + \beta \cdot [\, w \in \bigcap B \,],$$

where $\bigcap \varnothing = W$ and $[\cdot]$ is 1 if the bracketed condition holds and 0 otherwise. A belief state earns the separable score of its members, plus β at exactly those worlds where everything it contains is true. The premium is the value of an error-free record. One rationale is inferential: a belief state whose members are all true is one from which any amount of multi-premise deduction yields only truths. A body of belief can thus be valuable as a whole, which is the holist's thought. Dorst motivates Separability with an exam: "her score on the exam is the sum of her scores on each question" (8). Many exams also reward a perfect paper, and the record measure scores belief that way. The measure is purely alethic: $A_\beta(B, w)$ depends only on B and on which of its members are true at w.

It meets every premise of Dorst's §4.2. Adding a true p raises A_β by exactly T_p: if every member of B_0 was true at w, all still are. Adding a false p lowers it by $\mid F_p \mid$, or by $\mid F_p \mid + \beta$ if B_0 was error-free at w. So truth-directedness holds. Conservativeness can be imposed on the v_p. The measure does not depend on the current state, so Maximize Expected Accuracy has a single, sequence-independent verdict and Sequence-Irrelevance holds. Every belief state is an option, so Accuracy First holds. I add one premise that Dorst's framework requires anyway:

(Permission) If a belief state uniquely maximizes expected accuracy, it is rationally permitted.

Dorst's own gloss is that "rationality requires you to have an attitude that best promotes accuracy" (2). If the unique maximizer were impermissible, nothing would be permissible.

The case. Let $W = \{w_1, w_2, w_3\}$, $T_p = 1$ and $F_p = -2$ for every p (so $\tau = 2/3$), and $\beta = 3$. Write $\mathbf{1} = \{w_1\}$. At $C = (0.70, 0.15, 0.15)$ the unique optimum among all 2^8 belief states is $\{\mathbf{1}, \{w_1, w_2\}, \{w_1, w_3\}, W\}$, with expected accuracy 4.30 against a runner-up of 4.20. At $C' = (0.75, 0.01, 0.24)$ the unique optimum is $\{\{w_1, w_3\}, W\}$, with 4.94 against 4.75. Hand computa-

tion agrees with exhaustive enumeration in exact arithmetic.[2] So $\mathbf{1}$ is believed at credence 0.70 and not at credence 0.75. By Permission both optima are rational. Proposition 5 would then require $\tau_{\mathbf{1}} \leq 0.70$ and $\tau_{\mathbf{1}} \geq 0.75$, which is impossible.

This is not a knife-edge.

Theorem 2. Let $T > 0 > F$ with $T < -F$, let $\beta > 0$, let $D = T - F$ and $\tau = -F/D$, and put

$$c_0 = \frac{\beta - T - 2F}{2D + \beta}, \qquad c_1 = \frac{\beta - 2F}{2D + \beta}.$$

Then $\max(c_0, \tau) < c_1 < 1$. For all $c < c'$ in the interval $(\max(c_0, \tau), c_1)$, $\mathbf{1}$ belongs to the unique optimum at $(c, \frac{1-c}{2}, \frac{1-c}{2})$ and to no optimum at $(c', \varepsilon, 1 - c' - \varepsilon)$, for all sufficiently small $\varepsilon \geq 0$.

The proof is in the Appendix; it was also checked by enumeration for premiums from 0.01 to 1000. The mechanism is visible in the case. At C the residual 0.30 is spread over two worlds, and the hypothesis $\mathbf{1}$ is the best "core": believing everything it entails earns three positive separable terms and a premium on 0.70. At C' almost all the residual sits on w_3, so the disjunction $\{w_1, w_3\}$ is nearly certain. Its premium on 0.99 outweighs what the more specific hypothesis adds, so the agent hedges. As β shrinks the interval narrows (its length above τ is $\beta T/(D(2D + \beta))$), but it never closes.

The result instantiates Holguín's principle of Non-Monotonicity: "There can be two agents (or the same agent at different times) S_1 and S_2 such that: $C_{S_1}(p) < C_{S_2}(p)$, yet S_1 is rationally permitted to think that p while S_2 is not" (2022, §5). His argument against Lockeanism used this principle. The record measure shows that accuracy-first epistemology without Separability is not bound by monotonicity either.

4. Diagnosis

Where does the derivation fail? There are two gaps.

The quantifier gap. Truth-directedness gives, for each world w, some positive increment. The passage quoted in §2 then names one number $T_p^{B_0}$ as "the" increment. That moves from "for every w there is a t" to "there is a t for every w." Under the record measure the step fails. Adding the false $\mathbf{1}$ to $\{\{w_1, w_3\}\}$ costs 2 at w_2 but 5 at w_3, since only at w_3 was the prior state error-free. In expectation, the comparison that Proposition 4 makes between B and $B \setminus \{p\}$ becomes

$$EA_C(B) - EA_C(B \setminus \{p\}) = e_p - \beta\, C\left(\bigcap(B \setminus \{p\}) \setminus p\right).$$

Belief in p is therefore worth adding iff e_p exceeds a term fixed by the credence of the rest of the core outside p. That is a threshold on $C(p)$ that moves with credences in other propositions, not one fixed by values. What the step needs is:

[2] All 256 states over the eight propositions of a three-world space (including $\varnothing$ and W) were enumerated with rational arithmetic. A scan of credences on a grid of mesh $1/40$ found monotonicity violations for all six contingent propositions.

(ME) *Marginal extensionality.* For every B_0 and $p \notin B_0$, $A(B_0 \cup \{p\}, w) - A(B_0, w)$ depends on w only through whether $w \in p$.

This is not a consequence of A's being well-defined. It is a substantive claim: the marginal value of a belief may depend on the world only through that belief's own truth value.

The content gap. Grant ME. Proposition 4 then yields thresholds $\tau_p^{B^-}$ indexed to B^-, the rest of the belief state. The proof of Proposition 5 generalizes from "any" B^- to all rational states. That step requires $\tau_p^{B^-}$ not to vary with B^-. Sequence-Irrelevance cannot supply this, because B^- is not a sequence; it is the content of the other beliefs, which is exactly what the holist says matters. For any fixed measure, Maximize Expected Accuracy compares whole states, and its verdicts are already sequence-independent. Dorst himself observes that "no holist ever thought that the insight of their position is that the order in which you consider propositions affects what you ought to believe" (17). That is why a principle about order cannot exclude the holism of content. What the step needs is:

(BI) *Background-independence.* $T_p^{B_0}$ and $F_p^{B_0}$ do not depend on B_0, for states B_0 not containing the complement of p.

Together, the two premises are Separability.

Theorem 1. Let A be truth-directed. Consider only states containing no proposition together with its complement.[3] (a) A satisfies ME iff $A(B, w) = A(\varnothing, w) + \kappa(B) + \sum_{p \in B} J_p\,[w \in p]$ for some world-independent κ and constants $J_p > 0$, the sum running over the contingent $p \in B$. (b) A satisfies ME and BI iff κ is additive, that is, iff $A(B, w) = A(\varnothing, w) + \sum_{p \in B} v_p(w)$ for some proposition-dependent values v_p.

Since $A(\varnothing, w)$ is the same for every state, it never affects which states are optimal. Part (b) therefore says that the measures for which Dorst's steps are sound are exactly those covered by Corollary 2. The proof (Appendix) turns on one observation. Adding p and q in either order gives the same total, so $\Delta_p^B - \Delta_p^{B \cup \{q\}} = \Delta_q^B - \Delta_q^{B \cup \{p\}}$, where Δ_r^B is the increment of adding r to B. Under ME the left side is affine in $[w \in p]$ and the right side affine in $[w \in q]$. Unless q is p's complement or trivial, both sides must be constant. A linear-algebra computation over all 256 states on three worlds confirms (a), and it confirms that the complement proviso cannot be dropped.[4]

Part (a) shows what survives of holism under ME: a world-independent term $\kappa(B)$, a bonus or malus for a state's shape regardless of how the world is. If κ is not additive, Proposition 5 still fails. For example, a small bonus for deductively closed states, small enough to preserve truth-directedness, shifts the effective threshold of p according to whether adding p completes or breaks

[3] The proviso is needed, since there are ME measures in which the marginal value of p changes when its complement is believed. It is harmless here. Under Variable Conservativeness $\tau_p + \tau_{\neg p} > 1$, so such states are never optimal for separable measures (Dorst's Proposition 3), nor, by Theorem 3 below, for the record measure.

[4] The ME constraints form a homogeneous linear system on the 768 values $A(B, w)$. Each vector of a basis of its solution space was tested against the representation in (a), with numerical deviation below 10^{-14}.

closure.[5] Dorst can reject such a κ as non-alethic. But then the only holism his argument tolerates is none.

Theorem 1 concerns the derivation, not its conclusion. Some non-separable measures do have Lockean optima. A separable measure lowered on states that are never optimal is an example. So Separability is not necessary for Lockean outputs. What Theorem 1 shows is narrower and enough for my purpose: the separability-free route to Lockeanism does not exist, and the most natural alethic non-separable measure (Theorem 2) is not one of the lucky exceptions.

5. What holism delivers instead

The record measure does not leave belief unstructured. Its optima have an exact description. Write $\bigcap B$ for the *core* of B, and for each proposition X let

$$\Phi(X) = \sum_{p \supseteq X} \max(e_p, 0) + \beta\, C(X).$$

Theorem 3. For any finite W and any proposition-dependent values, $\max_B EA_C(B) = \max_X \Phi(X)$. A state B is optimal iff its core maximizes Φ, B contains every $p \supseteq \bigcap B$ with $e_p > 0$, and B contains no p with $e_p < 0$. In particular, when the values are uniform and C is generic, the unique optimum is

$$B^* = \{p \supseteq X^* : C(p) > \tau\},$$

where X^* maximizes Φ, and: (i) B^* is consistent iff $X^* \neq \varnothing$, and it is closed under single-premise entailment; (ii) X^* contains every world more probable than any world it contains; (iii) B^* is deductively closed iff $C(X^*) > \tau$, in which case it consists of exactly the consequences of X^*.

Call this *core-relative Lockeanism*. The agent selects a working hypothesis X^*, trading its probability (the premium term) against the number of above-threshold propositions it entails. She then believes those above-threshold propositions that the hypothesis entails. Lockean belief is the special case in which the best core is the intersection of all above-threshold propositions. By (ii) the working hypothesis is always a set of most probable worlds, which is the filtering structure guess-based theories posit for answers (Holguín 2022, §10). By (iii) closure is not guaranteed. On four worlds with $C = (0.62, 0.18, 0.10, 0.10)$ and the values of §3, the unique optimum believes $\{w_1, w_2\}$ and $\{w_1, w_3\}$ but not $\{w_1\}$. The theorem was checked against exhaustive enumeration on three and four worlds, with uniform and with randomly varying values.[6]

Three consequences matter for the debate.

First, accuracy-first epistemology is neutral between Lockean and guess-like belief. Separable value gives thresholds. Record value gives a best hypothesis and its probable consequences. Accuracy does not choose between them; a view about the value of an error-free body of belief does.

[5] With $T = 1$, $F = -2$ and a bonus of $1/2$ for closed states, the measure satisfies ME and truth-directedness. On three worlds, $\{w_1\}$ is believed at $(0.575, 0.2, 0.225)$ but not at $(0.65, 0, 0.35)$.

[6] Over 900 random instances with integer credences, the predicted and the computed optimal sets coincided in every case.

Second, the record measure captures both forms of non-monotonicity in the literature. In §3 the agent hedges when a disjunction becomes nearly certain. In Holguín's horse race, belief in A is lost when a rival overtakes it: told that A, B and C have chances 0.40, 0.35 and 0.25, one may think A wins; told 0.45, 0.50 and 0.05, one should think B wins and not A (2022, §5). A single context reproduces both verdicts. Let values depend only on how many horses a proposition mentions: $(T, F) = (1, -0.66)$ for one horse and $(0.65, -1.01)$ for two, which satisfies Dorst's Variable Conservativeness, and set $\beta = 0.19$. The unique optimum at the first credences is the set of consequences of "A wins". At the second it is the set of consequences of "B wins". The runner-up there is the inconsistent state that believes both A and B, which is exactly what a Lockean with these thresholds (about 0.40 for one horse) would believe.[7] So the premium does what Holguín says sophisticated Lockeanism cannot: it keeps thresholds low enough for extreme weakness while ruling out the incompatible beliefs those thresholds would license. The margins are small, though, and a random search suggests that values doing this occupy a narrow band. The result shows that non-monotonicity is available to accuracy-firsters. It does not show that the record measure is the right theory of thinking.

Third, Dorst's reduction of belief to credence (§5.1) survives in a weakened form. Under the record measure, belief can still be "read off" credence and contextual values, since B^* is a function of both. What fails is the Lockean form of the function. The reduction is thus independent of Lockeanism. A Lockean needs Separability as a further, substantive commitment.

6. The boundary: split-invariance

If holism can deliver closed and filtered belief, can it deliver the belief of knowledge-aimed theories? Goodman and Salow model belief as the evidential component of knowledge. On their probabilist proposal, "w is sufficiently more normal than v just in case the evidential probability of things being more normal than v, given that they are no more normal than w, is sufficiently high" (2023, 121). In continuous cases belief "is characterized by the highest posterior density region for the given threshold" (122). For a question with finitely many answers and threshold t, their definitions imply the following *probabilist normality rule*: answer v is ruled out iff the answers strictly more probable than v jointly have probability at least t.[8] This rule has a feature that no expected-value rule has:

(S) *Split-invariance.* If the strongest believed answer is a, then redistributing the probability of the other answers among them, keeping each below $P(a)$, leaves the strongest belief unchanged.

Consider a three-answer question $\{a, b, c\}$. Assign each disjunction of answers G a value T_G if true and F_G if false (the tautology has only T_W). Let the believed answer maximize $P(G)T_G + (1 - P(G))F_G$, with ties broken arbitrarily. Assume truth-directedness ($T_G > F_G$) and informa-

[7] Exhaustive enumeration over all 256 belief states, in exact arithmetic, with $T_W = 1$ (the value of believing W does not affect which states are optimal) and $\beta = 19/100$. Expected accuracies: 1.384 against 1.380 at the first credences, and 1.832 against 1.824 at the second.

[8] Belief rules out v iff some answer is sufficiently more normal than v (Goodman and Salow 2023, 112–113). The conditional probability in their definition is largest when w is a most probable answer, which yields the rule stated. Goodman and Salow have "no settled view" between this probabilist approach and a multi-dimensional one (2023, 123).

tiveness ($T_G > T_W$ for two-answer G). Add two weak conditions for an answer i: anti-scepticism, AS(i), which says that $\{i\}$ is believed whenever $P(i)$ is close enough to 1; and majority caution, C(i), which says that $\{i\}$ is never believed when $P(i) < 1/2$.

Theorem 4. No such expected-value rule satisfies S together with AS and C for two of the three answers. The premise set is minimal: dropping any one of S, expected-value maximization, truth-directedness, informativeness, AS, C, or the second answer leaves a satisfiable set.

The idea of the proof is this. Let x^* be the credence in a above which $\{a\}$ is always believed; C puts $x^* \geq 1/2$. Split-invariance spreads the verdict along the whole slice $P(a) = x^*$. At the slice's endpoints a two-answer disjunction is certain, so the value of believing $\{a\}$ there is at least $\max(T_{ab}, T_{ac})$. Just below x^*, where the residual is split evenly, some other answer is chosen, and by continuity it ties with $\{a\}$ at x^*. Every candidate leads to a contradiction except the complement $\{b, c\}$, and that candidate forces $T_{bc} > \max(T_{ab}, T_{ac})$. Running the same argument for b forces $T_{ac} > T_{bc}$. The independence models are in the Appendix and were machine-checked. The probabilist normality rule satisfies S, AS and C for every threshold $t \in (1/2, 1)$, so it is not an expected-value rule. The record measure, by contrast, is split-sensitive: at $(0.75, 0.125, 0.125)$ its strongest belief is $\{w_1\}$; at $(0.75, 0.01, 0.24)$ it is $\{w_1, w_3\}$.

Two calibrations. First, Theorem 4 concerns values assigned to answers, with a world-independent value for each false answer. Restricted to closed states, the record measure is of this form, because a closed state's false members at a world w outside its core are equal in number for every such w. The general state-level case is open. I conjecture that it also excludes split-invariance. Second, the theorem costs knowledge-aimed theorists little if they never sought an accuracy grounding. Goodman and Salow follow "the tradition in epistemology of thinking of belief as aiming at knowledge" while agreeing that "'believe' in English expresses something considerably weaker" (2023, 113 n. 21). What the theorem blocks is a hybrid: closed, knowledge-aimed belief with an accuracy-first foundation. The theorem also bears on Dorst's own proposal. He grants that Lockean belief cannot play the knowledge-aimed roles. He assigns them to "being sure, or being certain, or having credence 1, or taking for granted, or taking yourself to know—all the same (context-sensitive) state" (21). Read as a rule on a fixed credence function, "believe what has credence 1" satisfies S and C but not AS. That is the one escape Theorem 4 leaves open to a split-invariant attitude: give up anti-scepticism. It also differs in form from Wang's (2025) reported result that the stability theory of Leitgeb (2014) violates the norms implied by expected-utility binarization. Theorem 4 isolates a single checkable property, and it covers the normality rule. I have not been able to check whether the normality rule violates Wang's convexity norms.

7. Objections

The premium is not accuracy; it double-counts the conjunction. The premium is alethic, since it depends only on which believed propositions are true. It is not a disguised belief in the core: in the four-world case it accrues on $\{w_1\}$, which is not believed. And it respects strict truth-directedness, the only constraint Dorst places on holistic measures. The objector may insist that genuine accuracy is local, so that a belief's contribution depends only on its own truth. That insistence is ME, and by Theorem 1 it is Separability waiting to happen. To exclude the premium

on those grounds is to deny the holism that §4.2 set out to accommodate. The reply leaves one thing open: someone may find the premium unmotivated as a value. Then the dispute is about the value of an error-free record. That is my thesis.

Dorst meant only belief-dependence: values that vary with what else is believed, not with whether it is true. The text resists this reading: the step "assumes only that A_{B_0} is well-defined." Grant the reading anyway. ME then becomes an explicit premise, and the content gap remains. Under ME, Theorem 1(a) leaves a world-independent κ. If κ is non-additive, Proposition 5 fails; if it is additive, the measure is separable. On the charitable reading, then, Proposition 5 still does not follow from the stated premises, and the holism it accommodates is empty.

Sequence-Irrelevance should be read strongly, as requiring background-independent thresholds. So read, it is no longer about sequences. It is BI under another name, and Dorst's defence of it (renumbering the questions on an exam should not change the rational answers, 16) supports only the order reading. The strong reading also undercuts §4.2's conclusion. Showing that a holist must accept BI is just showing that the holist must accept Separability at the margin.

Variable Lockeanism lets thresholds shift with context; perhaps C and C' are different contexts. In Dorst's picture, context fixes the values T_p and F_p, and they fix the thresholds. In the counterexample the values and the premium are the same at C and C'; only the credences differ. If thresholds could move whenever credences move, variable Lockeanism would become the empty thesis Dorst himself guards against. He imagines the objection that "Give me any profile of beliefs you like, and it looks like I'll be able to give you a function from contexts-and-propositions to thresholds that fits it" (17), and he answers it by tying thresholds to values. Proposition 5 promises "belief-independent values" (16), and under the record measure no such values exist.

Maximize Expected Accuracy gives only a necessary condition; perhaps neither optimum is rational. Then rationality would be unsatisfiable at C and C'. Accuracy First would then have yielded a norm that no state can meet, which no accuracy-firster accepts. Permission is the minimal premise needed for the framework to issue verdicts at all.

The result is negative only; Dorst's picture survives. In part, yes. Belief remains a function of credence and contextual values, and Dorst's "best guess" gloss fits core-relative Lockeanism at least as well as Lockeanism. What does not survive is the claim that accuracy forces the Lockean form. That matters to Holguín's argument, which targets the Lockean form. Non-monotonicity tells against sophisticated Lockeanism, but not against accuracy-first epistemology as such.

8. Conclusion

I have shown three things. Dorst's derivation of Lockeanism without Separability is sound exactly for separable measures (Theorem 1). A natural holistic measure satisfying all his stated premises violates Proposition 5 for every positive premium (Theorem 2). That measure delivers core-relative Lockeanism, under which belief is consistent, filtered and non-monotone (Theorem 3). On the boundary, no truth-directed answer-level value yields split-invariant belief (Theorem 4). I have not shown that Lockeanism is false or that the premium is the right value. Nor have I settled the state-level case of Theorem 4 or whether any accuracy measure reproduces Holguín's horse race. What follows is a reallocation of the burden. The accuracy-first argument for the Lockean

thesis is exactly as strong as the case for Separability. The choice between thresholds and best hypotheses is a choice about whether a flawless body of belief is worth more than its parts.

Appendix: Proofs and checks

Proof of Theorem 1. Write $\Delta_p^B(w) = A(B \cup \{p\}, w) - A(B, w)$ for $p \notin B$. Under ME, $\Delta_p^B = F_p^B + J_p^B[w \in p]$ for contingent p, and Δ_p^B is constant for $p \in \{\varnothing, W\}$; truth-directedness gives $F_p^B < 0 < F_p^B + J_p^B$. (i) Let $p, q \notin B$ be distinct, with q not the complement of p. Computing $A(B \cup \{p, q\}, \cdot) - A(B, \cdot)$ in both orders gives $\Delta_p^B - \Delta_p^{B\cup\{q\}} = \Delta_q^B - \Delta_q^{B\cup\{p\}}$. The left side is $\alpha + (J_p^B - J_p^{B\cup\{q\}})[w \in p]$ and the right side is $\gamma + \delta[w \in q]$, or a constant if q is trivial. If p, q are contingent, distinct and not complementary, then $1, [w \in p], [w \in q]$ are linearly independent, so $J_p^{B\cup\{q\}} = J_p^B$. If q is trivial, the same conclusion follows. (ii) Building any B that contains neither p nor its complement from $\varnothing$ one member at a time, (i) gives $J_p^B = J_p^\varnothing =: J_p$. (iii) For B without complementary pairs, add its members in any order. Each increment is $F_p^{B_i} + J_p[w \in p]$, so $A(B, w) - A(\varnothing, w) = \kappa(B) + \sum_{p\in B} J_p[w \in p]$, where $\kappa(B)$ is the sum of the F-terms (and the constant increments). It is world-independent, and it is order-independent because the total is. The converse of (a) is immediate. For (b), BI says $F_p^B = \kappa(B \cup \{p\}) - \kappa(B)$ is independent of B, which holds iff $\kappa(B) = \kappa(\varnothing) + \sum_{p\in B} f_p$; then $v_p = f_p + J_p[w \in p]$. Conversely, separable measures satisfy ME and BI trivially. ∎

Proof of Theorem 3. $EA_C(B) = \sum_{p\in B} e_p + \beta C(\bigcap B)$. Every member of B includes $\bigcap B$, so $EA_C(B) \le \Phi(\bigcap B)$, with equality iff B contains every $p \supseteq \bigcap B$ with $e_p > 0$ and no p with $e_p < 0$. Conversely, given X, let $B_X = \{p \supseteq X : e_p > 0\}$ and $X' = \bigcap B_X \supseteq X$. The members of B_X are exactly the positive-e propositions including X', so $EA_C(B_X) = \Phi(X') \ge \Phi(X)$, because $C(X') \ge C(X)$. Hence $\max EA = \max \Phi$, and the optimality condition follows. For uniform values, upward closure holds because $p \subseteq q$ implies $C(q) \ge C(p)$. For (ii): if $u \in X^*$, $v \notin X^*$ and $C(v) > C(u)$, swapping u for v maps the supersets of X^* bijectively onto those of the new core without lowering any credence, and it raises $C(X^*)$; that contradicts maximality. For (iii): B^* is upward closed with $\bigcap B^* = X^*$, so it is closed under conjunction iff $X^* \in B^*$, iff $C(X^*) > \tau$. Consistency holds iff $\bigcap B^* \neq \varnothing$. ∎

Proof of Theorem 2. Write $e_p = C(p)D + F$; note $\tau > 1/2$. At $C_c = (c, \frac{1-c}{2}, \frac{1-c}{2})$ with $c > \tau$, the positive terms are $e_{\mathbf{1}}$, $e_{12} = e_{13}$ and $e_W = T$, and all others are negative (their credences are at most $1 - c < 1/2$). Then $\Phi(\mathbf{1}) - \Phi(W) = e_{\mathbf{1}} + 2e_{12} - \beta(1-c) = c(2D+\beta) + T + 2F - \beta$, which is positive iff $c > c_0$. Also $\Phi(\mathbf{1}) - \Phi(\{w_1, w_3\}) = e_{\mathbf{1}} + e_{12} - \beta\frac{1-c}{2} > \frac{1}{2}(\Phi(\mathbf{1}) - \Phi(W))$, which is positive. Next, $\Phi(\mathbf{1}) - \Phi(\{w_2\}) = e_{\mathbf{1}} + e_{12} + \beta\frac{3c-1}{2} > 0$ and $\Phi(\mathbf{1}) - \Phi(\{w_2, w_3\}) = e_{\mathbf{1}} + 2e_{12} + \beta(2c-1) > 0$; the cases of $\{w_1, w_2\}$ and $\{w_3\}$ are symmetric; and $\Phi(\mathbf{1}) - \Phi(\varnothing) = \beta c > 0$. By Theorem 3 the unique optimum has core $\mathbf{1}$ and contains $\mathbf{1}$. At $C' = (c', 0, 1-c')$ with $\tau < c'$: $e_{\mathbf{1}} = e_{12} = c'D + F > 0$ and $e_{13} = e_W = T$, so $\Phi(\{w_1, w_3\}) - \Phi(\mathbf{1}) = \beta(1-c') - 2e_{\mathbf{1}}$, which is positive iff $c' < c_1$, while $\Phi(\varnothing) = \Phi(\mathbf{1}) - \beta c'$. A state containing $\mathbf{1}$ has core $\mathbf{1}$ or $\varnothing$, so no optimum contains $\mathbf{1}$. These strict inequalities persist for small ε. Finally, $c_1 - c_0 = T/(2D+\beta)$, $c_1 - \tau = \beta T/(D(2D+\beta))$ and $1 - c_1 = 2T/(2D+\beta)$ are all positive. ∎

Proof of Theorem 4. Fix a and let $x^* = \inf\{x : \{a\} \text{ is chosen at every } P \text{ with } P(a) > x\}$. AS gives $x^* < 1$; C gives $x^* \ge 1/2$. For $y > x^*$, $\{a\}$ is optimal on the whole slice $P(a) = y$. By continuity,

its value $V = x^*T_a + (1-x^*)F_a$ is maximal on the slice $P(a) = x^*$; at the slice's endpoints this gives $V \geq T_{ab}$ and $V \geq T_{ac}$. There are $y_n \to x^*$ with $y_n \leq x^*$ such that $\{a\}$ is not chosen at the even point $E_{y_n} = (y_n, \frac{1-y_n}{2}, \frac{1-y_n}{2})$: for $y_n > 1/2$ this follows by S from a non-$\{a\}$ point on the slice, and for $y_n < 1/2$ by C. Some answer $G \neq \{a\}$ is chosen infinitely often, and its value at E_{x^*} equals V. Now $G = W$ gives $T_W = V \geq T_{ab} > T_W$. $G = \{a, b\}$ gives the value $\frac{1+x^*}{2}T_{ab} + \frac{1-x^*}{2}F_{ab} < T_{ab} \leq V$; $\{a, c\}$ is symmetric. $G = \{b\}$ has a strictly larger value at the endpoint $(x^*, 1-x^*, 0)$ than at E_{x^*}, so it would beat $\{a\}$ there; $\{c\}$ is symmetric. Hence $G = \{b, c\}$, whose value is constant on the slice, and $T_{bc} > (1-x^*)T_{bc} + x^*F_{bc} = V \geq \max(T_{ab}, T_{ac})$. The same argument for b yields $T_{ac} > T_{bc}$, a contradiction. Independence models: drop S, with size-only values $T = (1, 0.6, 0.5)$ for one-, two- and three-answer G, and $F = 0$; drop expected-value maximization, with the normality rule, $t \in (1/2, 1)$; drop AS, with $T = (0.1, 1, 0.9)$, $F = 0$; drop C, with $T = (1, 0.5, 0.4)$, $F = 0$; drop truth-directedness, with $T_{\{i\}} = 3$, $F_{\{i\}} = -2$, $T_G = F_G = 1$ for two-answer G, $T_W = 0.5$; drop informativeness, with $T_{\{i\}} = 1$, $F_{\{i\}} = -1$, $T_G = 0$, $F_G = -1$ for two-answer G, $T_W = 0.2$; keep AS and C for one answer only, with $(T, F) = (1, -1)$ for $\{a\}$, $(1, -10)$ for $\{b\}, \{c\}$, $(0.7, -10)$ for $\{a, b\}, \{a, c\}$, $(8, 0)$ for $\{b, c\}$, $T_W = 0.5$. Each model was verified on a grid of mesh $1/240$ to satisfy every remaining premise. ∎

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